Posterior Instance Injection Detector for Arbitrary-Oriented Object Detection From Optical Remote-Sensing Imagery

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***Abstract*— Arbitrary-oriented object detection (AOOD) from optical remote-sensing imagery has to correctly generate delicate oriented boundary box (OBB) and meanwhile identify their specific categories. However, how to make detectors learn the delicate parameters of OBBs, especially for the crucial orientation information, and identify object categories from complex backgrounds becomes a challenging task. Therefore, in this article, to explore a better way to guide the detector to learn specific categories and parametric information of OBBs, a novel one-stage anchor-free detector called posterior instance injection detector (PIIDet) is proposed for AOOD. First, as the anchor-free manner lacks prior information, an object-aware posterior guidance (OAPG) structure is proposed to generate specific-category instances used for conditioning on OBB prediction. This structure can assist the proposed PIIDet in better learning the relative parametric information of OBBs corresponding to their specific categories. Besides, to guarantee a high-quality injection of specific category instances, a new hierarchical feature fusion module (HFFM) is developed to establish a suitable multiscale feature mapping space. Second, considering the negative optimization of angle regression, which is caused by the boundary discontinuity of angular periods and sudden shifts of the relation between width and height in the training phase, a novel binary classification embedded angle regression space (BCE-RegSpace) is devised for providing continuous angle regression space and stable relation between the width and height. Finally, extensive experiments are executed on three AOOD benchmarks (e.g., DOTA, DIOR-R, and HRSC2016), and results proved that the proposed concise one-stage anchor-free PIIDet can reach the state-of-the-art (SOTA) performance and meanwhile have an impressive inference speed.**

***Index Terms*— Arbitrary-oriented object detection (AOOD), one-stage anchor-free detector, optical remote sensing.**

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I. Introduction

ARBITRARY-ORIENTED object detection (AOOD) is a challenging task for optical remote-sensing imagery because it requires not only learning several delicate parametric information mapping of oriented boundary box (OBB) from large-scale overlooking scenes, but also identifying their specific categories amidst complicated background. Because of its capability to enable precise object detection, AOOD has become a critical technology that is applied in various real-world applications, including urban planning, resource monitoring, wide-area rapid search and rescue, and military applications [1], [2], [3], [4], [5]. Essentially, the objective of AOOD is to utilize the deep neural network to construct a mapping function between the parametric information of OBBs and their corresponding specific categories from images, which is similar to the mapping paradigm of horizontal boundary box (HBB) in natural scenes, as shown in Fig. 1. However, unlike the natural scene depicted in Fig. 1(a) with several objects distributed on clear background, Fig. 1(b) presents the more challenging remote-sensing scenario due to the presence of complicated background interference and the significant variation in scale and shape among objects from different categories. In addition, when comparing HBBs with OBBs, the established mapping function can easily map absolute parametric information such as category, width, and height, because they would not rely on other parameters and have straightforward representations. Conversely, relative parametric information, such as the angle in OBBs, often necessitates the establishment of a reference system to achieve the mapping. For example, the angle may have different periodic definitions (e.g., 180° and 90°), meanwhile, the value of the angle may be determined by either the long or the short side. This variability makes it more challenging to learn the mapping function for relative parametric information. To this end, the incorporation of better reference systems and guiding information is crucial. Based on this consideration, under the complex remote-sensing scene, constructing an effective mapping function for assisting the detectors in learning the precise parameters of OBBs, especially for relative angle parameters, and accurately identifying specific categories against complicated backgrounds becomes a formidable task.

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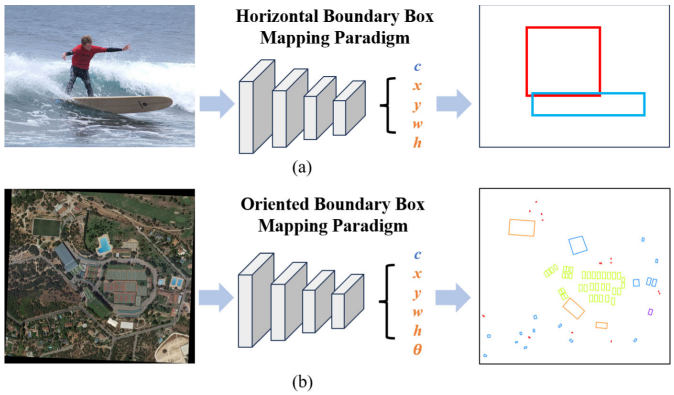


Fig. 1. Comparison of two types of boundary box mapping paradigms. (a) Horizontal boundary box mapping paradigm in natural scenes. (b) OBB mapping paradigm in remote-sensing scenes. c denotes the category; x, y denote position coordinates; w, h, and θ represent object width, height, and angle, respectively. Different colors represent different categories on the right of the figure.

To cope with these challenges, numerous researchers have dedicated great efforts. At present, the mainstream solution for tackling the characteristic of ever-changing object orientation revolves around the improvement of anchor-based detectors [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16]. The main reason for this is that anchor-based detectors employ a mechanism involving preset anchors, enabling the placement of a wide range of carefully designed rotating anchors in advance. It is obvious that these anchors can be effectively matched with the objects exhibiting diverse orientations. For instance, Zhang et al. [8] set six different angles, that is, 0, $\pi/6$, $\pi/3$, $\pi/2$, $2\pi/3$, and $5\pi/6$ to rotate the anchors into different orientations, which helps match almost ships. And then, Yang et al. [6], [7] empirically preset the orientation of each anchor with an interval of 15°, aiming to match arbitrary-oriented objects as much as possible. Despite this progress, discrete anchor design based on manual prior knowledge is limited in its ability to adapt to continuous orientation changes. Furthermore, if considering factors like scales, aspect ratios, and angles simultaneously in manual anchor designs, especially in the context of significant scale variation of remote-sensing objects, it would result in an excessively dense layout of anchors. As an example, Yang et al. [7] allocated 126 anchors with varying scales, angles, and aspect ratios for each sliding position of the feature pyramid. It is undeniable that the preset anchors have brought significant performance gain, but this is often restricted by specific datasets which would lead to massive computation and memory requirements. Recently, some studies [13], [17], [18], [19] committed to automatically generating high-quality anchors for aligning the oriented objects by adjusting a set of manual preset anchors. Although these approaches reduce the number of anchors, they may lead to a decrease in inference efficiency. Either way, the performance of AOOD is still dominated by empirically designed anchors.

Considering the limitations of anchor-based detectors, some scholars have begun to explore anchor-free detectors [20], [21] for AOOD. They considered the shape or orientation of objects to formulate label assignment rules for keypoints, such as 2-D oriented Gaussian center prior weights [22] and specified corner points [23], which employ the manual prior modeling to be more in line with the characteristics of the object's shape and orientation. Nonetheless, they still suffered from the impact of subjective error caused by manually formulated rules, as achieving a perfect fit for all arbitrary-oriented objects through subjective experience is too hard to guarantee. Regarding the aforementioned hand-designed rules based on human knowledge, they are referred to as prior guidance.

On the contrary, instead of designing prior guidance driven by subjective experience, several attempts [24], [25], [26] have been made to develop feature refinement approaches to promote the performance of anchor-free detectors on the AOOD task. The reason is that these anchor-free detectors are built upon a pixel-level prediction manner. This causes them to request a higher demand for pixel-level semantic feature representation that describes the relationships between the pixels of objects and their surrounding pixels, especially for arbitrary-oriented objects. For example, Lin et al. [24] built an interacting embranchment black using a self-attention module to obtain the features that come from both classification and regression branches and provide more practical features for relative angle parameter prediction. Then, to improve the AOOD performance of anchor-free detectors, Dai et al. [25] aggregated contextual information by a spatial feature refinement block and refined features by embedding them in the feature pyramid network. Moreover, Li et al. [26] proposed a crisscross attention feature pyramid network to fully exploit the contextual information of each pixel and its neighbors, which can help solve the severe shape ambiguity problems faced by anchor-free methods. However, despite this progress, the AOOD performance of anchor-free detectors without prior guidance still cannot surpass these detectors with prior guidance. In fact, since the AOOD requires the relative angle parameter of objects to be provided, the guidance used to assist detectors in perceiving the shape to understand the relative angle of objects is crucial for improving the performance of AOOD, which is why anchor-based detectors can achieve great performance gain from predefined anchors. As a result, in the anchor-free framework, the lack of auxiliary guidance for relative angle parameter prediction is the essential reason for hindering the performance improvement of AOOD. To tackle this issue, we first proposed the concept of posterior guidance, that is, developing learnable auxiliary parameters to guide the network in perceiving the object shape to understand relative angle information. Formally, these learnable auxiliary parameters can be optimized by the network itself, and then be coordinated with the learning of OBB mapping parameters, which circumvents the subjective experience errors that may arise from hand-formulated rules and is a more flexible and promising guidance approach than prior guidance.

In addition, apart from an effective guidance approach used to promote the mapping paradigm of OBBs, the modeling of OBB mapping parameters is a decision-level influencing factor, as the relative parametric information, that is, the angle is directly relevant to the quality of object localization. Previous works about the design of OBB mapping parameters mainly replied on the angle and quadrangle [7], [10], [12], [16], [17],

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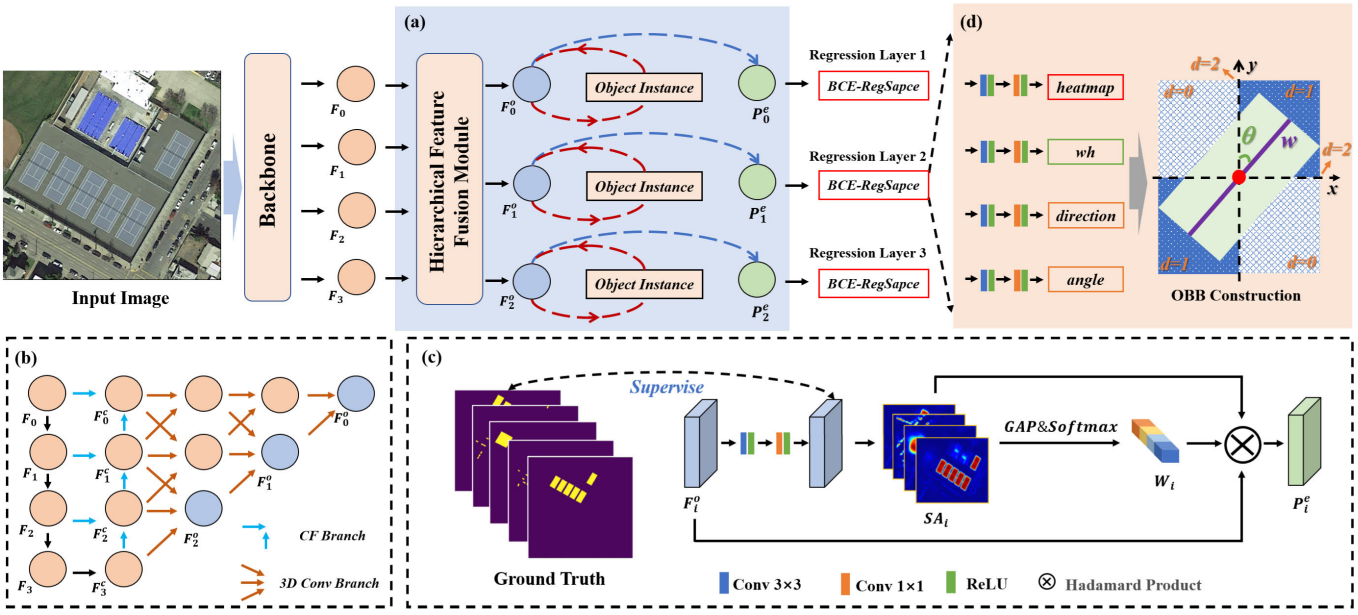


Fig. 2. Overall framework of our proposed PIIDet. (a) OAPG structure. (b) and (c) HFFM and OA-PII, respectively, which are implementation details of (a). (d) Proposed BCE-RegSpace OBB mapping parameter. In addition, the meaning of symbols is explained. $F_i, i \in [0, 3]$ denote the multiscale feature maps obtained from the output of the backbone network; $F_i^c, i \in [0, 3]$ denote the multiscale feature maps that are further fused by the CF branch; $F_i^o, i \in [0, 2]$ represent the three high-level feature maps that is processed by the HFFM; $P_i^e, i \in [0, 2]$ represent the three output feature maps of the OA-PII module. In addition, SA_i and W_i, respectively, represent the spatial attention map and the class weight coefficients.

[18], [19], [27]. Although these methods achieved promising results, they still encountered the issue of angle regression discontinuity, which leads to producing poor OBBs and affects convergence stability during training. To this end, Yang and Yan [28] transformed the angular prediction from a continuous regression into a discrete classification, which forms a more structured representation of relative angle information. However, the discrete classification would bring the accuracy loss for continuous angle changes. Therefore, they designed a circular smooth label technique with a window function, where the window function can ensure continuity at the boundary and alleviate accuracy loss. And, Zhu et al. [29] proposed a novel approach in which they embed the relative angle information using 2-D periodic vectors with different periodicities, instead of directly regressing the angle of OBBs. In this way, the 2-D periodic vectors can be continuous as the orientation changes. Although current methods have partially mitigated the issue of angle regression discontinuity by transforming the OBB representation, they still did not consider the essential reason for issues such as the boundary discontinuity of angular periods and sudden shifts of the relation between width and height. It needs to further achieve more reasonable angle regression space. In addition, current approaches often involve multiple split components to establish the mapping parameters of OBBs, which requires complex postprocessing.

In this article, to explore a better guidance method for mapping function and establish a reasonable OBB mapping parameter within a continuous angle regression space, an effective solution, named posterior instance injection detector (PIIDet), is proposed based on the concise anchor-free one-stage framework. It aims to generate delicate OBBs, meanwhile accurately identifying their specific categories without the need for prior knowledge. The overall framework is shown in Fig. 2. First of all, an object-aware posterior guidance (OAPG) is proposed to generate learnable instance description coefficients that are conditioned on the OBB parameter mapping function, as shown in Fig. 2(a). Notably, whether the learnable instance description coefficients are accurate enough is crucial for supporting the guiding process. Consequently, following our previous study [30], a powerful hierarchical feature fusion module (HFFM), is developed for formulating a suitable multiscale feature mapping space for assisting the generation of learnable posterior instance description coefficients, as shown in Fig. 2(b). Based on the powerful multiscale feature mapping space, a class-wise instance prediction branch is applied for generating the learnable instance description coefficients that comprehensively reflect the width, height, position, and orientation of objects. Then, the learned instance description coefficients are considered as posterior guidance information to condition on the output of the HFFM from class and spatial dimensions, as illustrated in Fig. 2(c). Second, considering the existing issues with OBB mapping parameters, such as angle regression discontinuity and sudden shifts in the relationship between width and height, an elegant solution is proposed to set up a stable angle regression space and corresponding OBB mapping parameter, called binary classification embedded angle regression space (BCE-RegSpace). In detail, following the character of quadrant symmetry, the original angular period is decoupled into two symmetrical parts, as shown in Fig. 2(d). Then, with the guidance of embedded binary classification for two symmetrical parts, the relative angle parameters would be predicted within a continuous angle regression space. Finally, extensive experiments were conducted on three public remote-sensing datasets. Compared with other methods, our proposed PIIDet achieved state-of-the-art (SOTA) performance, with

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80.21% mAP on DOTA [16], 70.35% mAP on DIOR-R [31], and 90.64% AP on HRSC2016 [32], meanwhile maintaining impressive inference efficiency.

In summary, the main contributions can be summarized in the following four points.

1) To our knowledge, we are the first to introduce posterior instance injection into the anchor-free architecture for improving the performance of AOOD. Accordingly, a novel one-stage anchor-free detector called PIIDet is proposed for AOOD, which can achieve the tradeoff between accuracy and efficiency.
2) A novel OAPG structure is proposed to establish an effective posterior guidance method for AOOD, which can condition the learnable instance description coefficients on the prediction of OBB mapping parameters. Moreover, it not only suppresses complex background interference, but also promotes the prediction of absolute and relative parameters.
3) To facilitate the prediction of learnable instance description coefficients, an HFFM was designed to hierarchically fuse adjacent features, which can construct an effective multiscale semantic feature mapping space to integrate comprehensive semantic information.
4) Building upon the quadrant symmetry character of the angle period, a new BCE-RegSpace is devised to provide the rational and stable angle regression space that can address the issue of angle regression discontinuity. Meanwhile, the embedded binary classification can avoid the angle sudden caused by the exchange of the relation between the width and height.

II. Related Work

A. AOOD in Remote-Sensing Field

With the improvement of optical remote-sensing image spatial resolution, the AOOD task has become a hot research topic in the remote-sensing field, since it can provide more delicate object localization than HBBs. Currently, a lot of efforts have been made such as rotated anchor design [8], [10], [12], [13], [16], [33], [34], [35], [36], sample assignment [22], [37], [38], loss function design [14], [15], [39], [40], and improved OBB mapping parameters [16], [24], [27], [41], [42], [43], [44]. For instance, Zhu et al. [34] added multiple hand-designed angles to control the orientation of anchors to cover the oriented objects effectively. Moreover, Han et al. [12] proposed a rotation-equivariant detector (ReDet) to explicitly encode rotation equivariance and rotation invariance and then set 15 anchors per location of each pyramid level for better matching arbitrary-oriented objects. Regarding the sample assignment, Li et al. [37] designed the dynamic reppoints learning to evaluate the spatial and feature alignment ability of the objects and introduced classification scores into the above learning process. This allows for the selection of samples that perform well in both the classification and regression tasks. For loss function design, Yang et al. [15], [39] made great progress such as a novel regression loss based on Gaussian Wasserstein distance and IoU-smooth-L1 loss to tackle the discontinuity problem of angle period boundary. In addition, some studies [24], [28], [29], [42], [45] translated angle prediction parameters to other mapping parameters (e.g., 2-D vectors [29], gliding vertex [24], and object axis [42]) to promote detection accuracy. As mentioned above, the current OBB detectors are almost focused on the feature representation and regression methods around anchor design, and there is relatively little research that analyzes the mechanism of the mapping function of OBB parametric information. Meanwhile, it can be observed that anchor-based methods have always dominated AOOD. Despite some researchers [23], [46], [47], [48], [49], [50] beginning to focus on anchor-free detectors, the upper bound of performance is always lower than anchor-based methods. Accordingly, in this work, we concentrate on the mechanism of posterior guidance to explore the potential of anchor-free detectors, aiming to significantly improve the performance of anchor-free detectors for AOOD while ensuring efficiency and simplicity.

B. Attention Mechanism

Recently, the attention mechanism has made great progress in the computer vision field, such as object detection [51], [52], [53], classification [54], [55], [56], and semantic segmentation [57], [58]. The essence of the attention mechanism is to guide the neural network to focus more on regions of interest while reducing the importance of irrelevant areas through appropriate weighting. For example, Hu et al. [56] proposed a squeeze-and-excitation network for image classification tasks, which adaptively reweights channel-wise feature responses by explicitly modeling interdependencies between channels. In addition, Tao et al. [58] presented an efficient hierarchical multiscale attention mechanism that helps with both class confusion and fine detail for semantic segmentation. In the object detection task, Ming et al. [59] designed a polarization attention module to generate different feature pyramids for classification and regression tasks, aiming to reduce noise interference and suppress the influence of irrelevant high activation regions. As mentioned above, it is evident that the attention mechanism can effectively enhance feature representation to help different models better predict in labeling mapping space, boundary boxing parametric mapping space, and pixel-level semantic mapping space. Besides, some studies [60], [61] also attended to make use of supervision information, that is, predicting the attention map by ground truths, which is mainly adopted on salient object detection tasks [61]. Inspired by it, Yang et al. [15] developed the concept of instance-level noise and designed a novel instance-level denoising module in a feature map, which was realized by supervised segmentation whose ground truth is approximately obtained by the bounding box in object detection. Moreover, Xu et al. [62] proposed a hierarchical semantic propagation framework to boost object detection performance in remote-sensing imagery, where the instance-aware semantic outputs can be propagated to suppress noisy background information, and enable a focus on the foreground information in the generation of horizontal region proposals. In these works, it has been preliminarily proved that this supervision-guided attention mechanism is helpful for the parametric information mapping of object detection.

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Besides, the instance description seems to be a more promising method because it can represent object categories, positions, and even shapes [63]. Therefore, in our work, we further explore the attention mechanism for assisting in promoting the OBB parametric mapping under the anchor-free architecture.

C. Orientation Estimation

OBB provides a more accurate object description than HBB, especially for remote-sensing objects with arbitrary orientation and densely arranged because it can accurately predict boundary boxes that tightly enclose the object boundary and prevent filtering out positive samples by nonmaximum suppression (NMS). However, compared with HBB-based detectors [64], [65], [66], the prediction of OBB needs to learn extra relative orientation information. Earlier, the commonly used orientation estimation methods [7], [15], [17] adopted the regression-based angle prediction, which includes two types of angle periods. First, for a 90° angular range, the angle is defined by the acute angle between the certain side of OBBs and the x-axis. And, the 180°-based definition uses the angle between the long side of the OBBs and the x-axis. Although the regression-based angle prediction has achieved great progress, these methods suffer from discontinuous angle boundary problems [28], which causes unstable model training at periodic boundaries. To tackle it, there are researchers focusing on better OBB mapping parameters to estimate orientation, such as vectors, more corner points, offsets of points, and so on. For example, Wang et al. [67] replaced rotated boundary box prediction by leveraging a segmentation task, which is easier to train and yields more accurate detection results. Moreover, Xu et al. [27] transformed the corner point regression into four relative gliding offsets on each corresponding side, avoiding the confusion issue of sequential label points for oriented objects. Recently, Yu et al. [68] proposed a novel angle-free strategy named bounding box projection (BBP), which derived OBB mapping parameters by designing special geometrical relationships among vectors without relying on angles for AOOD. Although the above methods have, to some extent, solved the problem of boundary discontinuity, there still exist problems such as accuracy loss and complex translation formulas. Thus, how to construct effective and concise OBB mapping parameters is worth exploring.

III. Proposed Methodology

In this section, Section III-A illustrates the overall framework of PIIDet. Then, the proposed OAPG is introduced in Section III-B. Moreover, a novel OBB mapping parameter called BCE-RegSpace is presented in Section III-C. Finally, the details of the loss function are elaborated in Section III-D.

A. Overall Framework

The overall framework of our proposed PIIDet is shown in Fig. 2. First, the backbone network extracts multiple scale feature maps $F_i, i \in [0, 3]$ from input images. Then, the OAPG structure is developed to generate learnable instance description coefficients for assisting in perceiving the relative angle parameter of objects from the comprehensive shape of instances. In the context of OAPG, to build an effective multiscale semantic feature mapping space to make sure the accuracy of learnable instance description coefficients, we followed our previous idea of the CSFF module in [30] and newly designed the HFFM to fuse the adjacent outputs of backbone to get three high-level semantic feature maps $F_i^o, i \in [0, 2]$. Next, the object-aware posterior instance injection (OA-PII) module introduces a class-wise instance prediction branch to generate learnable instance description coefficients. After that, these learnable instance description coefficients, viewed as the posterior information, are injected into the outputs of the HFFM through spatial and class dimensions. The goal is to guide the feature maps F_i^o to further assist the model in perceiving the comprehensive semantic instance information to achieve the prediction of relative angle parameters. Besides, the hierarchical semantic feature fusion derived from HFFM not only can enhance the fine instance description, but also reduce redundant background noises. In the detection heads of OBB mapping parameters, a concise, yet effective BCE-RegSpace is proposed for estimating the orientation, which decouples the discontinuous angle regression period by the guidance of embedded binary classification. Then, apart from the orientation of objects, the center point and size of objects are also predicted in the detection heads. Lastly, to construct the multiscale regression layers (MSRLs), three scale feature maps $P_i^e, i \in [0, 2]$ with $1/4$, $1/8$, and $1/16$ downsampling of the original image are applied for achieving the prediction of remote-sensing objects with various scales.

B. Object-Aware Posterior Guidance

To clarify motivation, the guidance methods used to condition the OBB parameter mapping function are briefly discussed. Formally, the input remote-sensing imagery can be defined as X, and the mapping parameter of OBBs is defined as Y. As mentioned in Section I, the aim in AOOD is to learn the parametric information mapping function $f : X \to Y$. And then, the prior guidance $f_{\text{prior}} : X \to Y$ and posterior guidance $f_{\text{posterior}} : X \to Y$ are formulated as follows:

$$f_{\text{prior}} = g(x) \Rightarrow g(x|a) \tag{1}$$

$$f_{\text{posterior}} = (g \circ h)(x) \Rightarrow g(x|h(x))h(x) \tag{2}$$

where $x \in X$; g denotes the mapping function from X to Y; h denotes the mapping function from X to learnable auxiliary parameter P, that is, $P = h(x)$. As shown in (1), a indicates the hand-designed prior information. It is evident that a directly affects the parametric information mapping f_{prior}, resulting in subjective experience dominating the mapping process. On the contrary to prior guidance $f_{\text{prior}} : X \to Y$, the learnable auxiliary parameter $p \in P$ is obtained by feeding x in mapping function h, and then the parametric information mapping process conditions on $h(x)$ in the posterior guidance $f_{\text{posterior}} : X \to Y$ of (2). The entire mapping process is optimized by the network itself, leading to a self-sustaining conditional OBB refinement. In comparison, the posterior guidance is obviously more promising than prior guidance, but similarly, the prediction quality of learnable auxiliary

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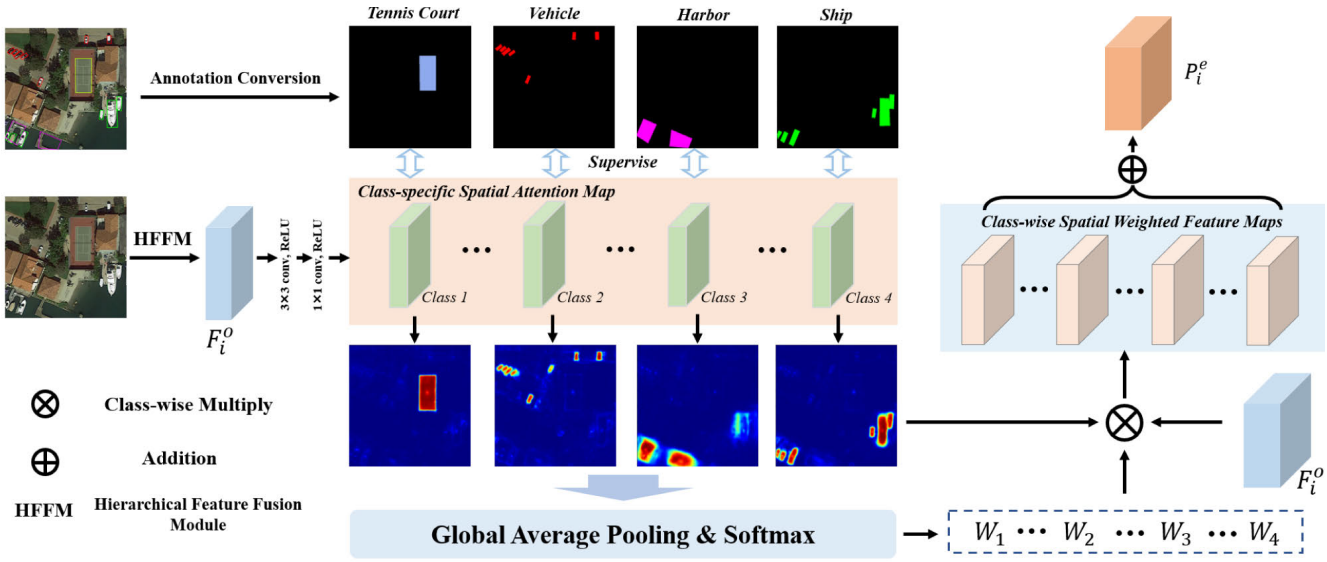


Fig. 3. Implementation details of the OA-PII module. Here, F_i^o represents the output feature map of the HFFM; P_i^e represents the feature map that is processed by the OA-PII module.

parameters also affects the effectiveness of posterior guidance. Specifically, some methods similar to posterior guidance [6], [46], [69] solely considered the foreground objects without specific categories and neglected the impact of comprehensive semantic description for guidance effectiveness, which causes that they are hard to learn the absolute or relative parametric information about multicategories and multiscale remote-sensing objects.

Consequently, inspired by the posterior guidance and to set up an effective guidance method to holistically promote the prediction of absolute parameters (e.g., category, width, and height) and relative parameters (e.g., angle), an OAPG structure is proposed, which includes the HFFM and OA-PII. First, in the OA-PII, a class-wise instance prediction branch is introduced to generate learnable instance description coefficients that are used for describing comprehensive shape information. Due to the good description of shape information, these instance description coefficients can assist the network in understanding absolute and relative parameter mappings. To supervise this learning process, the pixel-level quadrilateral mask derived from ground truth is viewed as a supervised signal, as shown in the upper side of Fig. 3. Notably, under the remote-sensing scenes, predicting fine object instances with various scales and orientations in complex backgrounds is challenging. It is urgent to ensure the quality of learnable instance description coefficients to introduce the multiscale compound semantic fusion. As a result, the HFFM is designed to construct an effective multiscale semantic feature mapping space. As shown in Fig. 2(b), four different scale feature maps $F_i, i \in [0, 3]$ are obtained from the backbone. Next, we first use the compensatory fusion module [30] to fuse F_i by the bottom-up way to obtain $F_i^c, i \in [0, 3]$, as follows:

$$F_i^c = \begin{cases} \mathrm{CF}\left(F_i, \mathrm{Upsample}\left(F_{i+1}^c\right)\right), & i \in [0, 2] \\ F_i, & i = 3 \end{cases} \tag{3}$$

where CF is the compensatory fusion module [30] and Upsample represents upsampling operation ($\times 2$). Then, the 3-D convolution [30] is applied on F_i^c to fuse adjacent two or three features three times, as shown in orange lines of Fig. 2(b). Lastly, multiscale feature outputs $F_i^o, i \in [0, 2]$ (i.e., three high-level feature maps) are obtained as shown in the blue circle of Fig. 2(b).

After possessing the multiscale semantic feature mapping space, as shown in Fig. 3, the 3×3 and 1×1 convolutions with rectified linear unit (ReLU) activation functions are applied on the OA-PII module to construct the class-wise instance prediction branch. Subsequently, the predicted results whose dimension is $h_i \times w_i \times C$ are injected into the outputs F_i^o of HFFM by class and spatial dimensions, where h_i and w_i are sizes of the ith feature layer and C denotes the number of categories. In detail, considering multiple categories are usually involved in complex remote-sensing scenes, it is necessary to decouple different categories to embody the category information. Therefore, each channel of predicted results can be regarded as a class-specific spatial attention map, that is, SA_i^c that reflects the comprehensive instance shape of the cth category objects. These spatial attention maps, respectively, condition the outputs F_i^o of the HFFM, which not only assists in mapping absolute and relative parameters, but also focuses on foreground areas. Furthermore, the class weight coefficients are calculated among different categories by first applying global average pooling on SA_i^c and then performing the softmax function on category dimension, the details are illustrated as follows:

$$W_i^c = \frac{\exp\left(\mathrm{GAP}\left(\mathrm{SA}_i^c\right)\right)}{\sum_c^C \exp\left(\mathrm{GAP}\left(\mathrm{SA}_i^c\right)\right)} \tag{4}$$

where W_i^c denotes the weight of the cth category, exp represents the exponential function based on natural constant e, and GAP represents the global average pooling. The channel weight W_i^c can adjust the relation among these categories, that is, weakening the impact of nonexistent categories. Finally, W_i^c and SA_i^c are processed by weighting the outputs F_i^o of the HFFM for obtaining the class-wise spatial weighted feature

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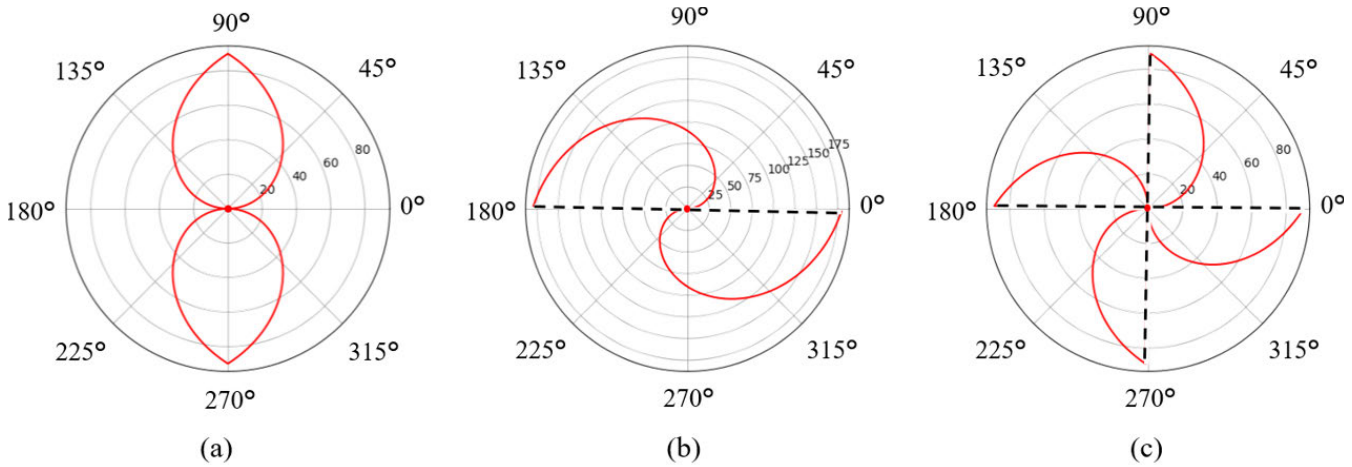


Fig. 4. Comparison of three OBB mapping parameters that is presented on the polar coordinate system. In detail, regarding (ρ, θ), ρ denotes the predicted angle and θ denotes the real angle. (a) Our designed BCE-RegSpace method. (b) 180°-base angle regression method. (c) 90°-regression methods. Notably, the red curves represent the trend of the predicted angle changing with the real angle; the black dashed lines indicate sudden changes in the predicted angle.

maps. Lastly, an addition operation is executed on these class-wise spatial weighted feature maps to obtain $P_i^e, i \in [0, 2]$, as shown in the right side of Fig. 3.

C. Binary Classification Embedded Angle Regression Space

As for traditional HBB mapping parameters, (x, y, w, h) are used for locating remote-sensing objects, where (x, y), w, and h represent the center point coordinates, width, and height of the objects. For the AOOD task, an extra parameter, for example, angle needs to be introduced to establish the OBB mapping parameters. Formally, most OBB detectors directly regress the angle θ, that is, (x, y, w, h, θ), which can be further categorized into two classical types. The first type involves regressing the angle θ in the 90° angular range, that is, $\theta \in [0, 90°)$, where θ denotes the acute angle between positive axis of the x-axis and one side of the OBB. And, the angle θ of the second type is determined by the long side and positive axis of the x-axis, whose angular range is $\theta \in [0, 180°)$. Although intuitive angle regression is reasonable, there exists the issue of angle regression discontinuity. The reasons contributing to this issue are mainly in two aspects. First, the predicted angle value often undergoes sudden changes at the angle periodic boundary. Concretely, as shown in Fig. 4, (ρ, θ) of the polar coordinate system is employed for representing the periodic change of the predicted angle with the real angle. Here, the red curve represents the trend of the predicted angle changing with the real angle. As shown in the black dashed lines of Fig. 4(b) and (c), it is obvious that the predicted angle changes sharply when the real angle changes slightly at the 90° or 180° angular periodic boundary, which would result in a sudden surge in the loss value of model during training phase. Second, in situations where there is a sudden shift in the relation between width and height due to inaccurate regression of absolute width and height parameters, it would impact the reference side of angle determination, resulting in a sharp change in the angle. As mentioned above, these two aspects both result in sudden changes in the loss value of angle prediction to complicate the accurate mapping of OBBs and hinder the overall model training process.

Considering the issues, we revisited the above angle regression methods and realized that the essence cause lies in the symmetry of the angular period because the relation between height and width used to determine the angle would exchange when crossing the axis of symmetry (e.g., x-axis or y-axis), which would cause the anomalous angle change. To solve it, a simple, yet effective OBB mapping parameter called BCE-RegSpace is designed, as shown in Fig. 2(d). First, a binary classification d is introduced to decouple the whole angle regression period into two directions by the character of symmetry, and then the relative angle θ [i.e., (x, y, w, h, θ, d)] is further predicted. In this way, the decoupled angle prediction effectively avoids the angle change discontinuity while increasing little computation. Concretely, the real angle θ' of OBBs is established by integrating the prediction of two branches. It is noteworthy that the long side of OBBs, that is, the width is considered the reference side for angle definition and the center point of OBBs is regarded as the origin of the coordinate system, as presented in the purple line and red point of Fig. 2(d). In the direction classification branch, three directions are defined according to the quadrant where the long side is located. When the long side locates (1, 3), (2, 4) quadrants, the direction d should be predicted as 0 and 1, respectively. In addition, to avoid the indistinct classification when the long side is parallel to the x-axis or y-axis, the direction d is predicted as 2. Moreover, the long side would be jointly constrained by the width and height regression and direction classification tasks, aiming to ensure the stability of the reference side selection. For the angle regression branch, no matter what the direction belongs to, the relative angle θ is determined by the long side of the object and the positive axis of the y-axis, ranging from 0° to 90°. The calculation formula is as follows:

$$\theta = \arccos\left(\frac{|y_b - y_a|}{\sqrt{(x_a - x_b)^2 + (y_a - y_b)^2}}\right) \tag{5}$$

where (x_a, y_a) and (x_b, y_b) represent the coordinates of the two endpoints of the long side. Based on the above definition, it can effectively avoid the angle regression discontinuity.

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As shown in Fig. 4(a), the regression of angle would not have an abrupt angle shift. In inference phase, the real angle θ' can be calculated by following simple:

$$\theta' = \begin{cases} 90 - \theta, & d = 0 \\ 90 + \theta, & d = 1 \\ \theta, & d = 2. \end{cases} \tag{6}$$

Moreover, in the training phase, due to the direction branch directly influencing the accuracy of OBB prediction, we adjust the loss weight among hybrid tasks and it would be further illustrated in ablation experiments.

D. Loss Functions

Given an input image $I \in R^{W \times H \times 3}$ of width W and height H, there are six branches to be predicted for training the proposed PIIDet, which include the center point (x_p, y_p), the offsets of center point (o_x, o_y), width and height (w, h), binary classification d, relative angle θ, and object instance m. Specifically, the overall loss function can be expressed as follows:

$$L = L_{ct} + L_m + L_{\text{off}} + \lambda_{wh} L_{wh} + \lambda_d L_d + \lambda_a L_{\text{angle}}. \tag{7}$$

If not specified, λ_{wh}, λ_a, and λ_d are set as 1, 0.2 and 1.

First, to locate the objects by predicting their center points, we followed the setting of CenterNet [20] and applied a size-adaptive Gaussian kernel

$$Y_{xyc} = \exp\left(-\frac{(x - x_g)^2 + (y - y_g)^2}{2\sigma_p^2}\right)$$

on each ground-truth center point (x_g, y_g), that is, positive sample, where

$$\sigma_p = \frac{(2 \times \max(w, h) + 1)}{6}.$$

Then, the pixel-wise classification loss L_{ct} is shown below

$$L_{ct} = \frac{-1}{N} \sum_{xyc} \begin{cases} \left(1 - \hat{Y}_{xyc}\right)^{\alpha} \log\left(\hat{Y}_{xyc}\right), & \text{if} Y_{xyc} = 1 \\ \left(1 - Y_{xyc}\right)^{\beta} \left(\hat{Y}_{xyc}\right)^{\alpha}, & \\ \log\left(1 - \hat{Y}_{xyc}\right), & \text{otherwise} \end{cases} \tag{8}$$

In (8), α and β are the weight adjustment parameters that are set as 2 and 4, respectively; $\hat{Y} \in [0, 1]^{(W/R) \times (H/R) \times C}$ is the prediction result; R is the downsampling stride and C is the number of categories; x and y denote the coordinate point in corresponding heatmap; c is certain category; and N is the number of center points.

Second, to make up for the discretization error caused by downsampling, L_{off} of offset $O_i = (o_i^x, o_i^y)$ is predicted by the following equation:

$$L_{\text{off}} = \frac{1}{N} \sum_{i}^{N} \left| O_i - \left(X_i^g - \lfloor X_i^g \rfloor\right) \right| \tag{9}$$

where i represents the center point of the ith ground truth and X_i^g represents the results by downsampling original center point coordinate (x_i^o, y_i^o), that is, $X_i^g = ((x_i^o/R), (y_i^o/R))$.

Third, to achieve the BCE-RegSpace-based OBB mapping parameter, it is necessary to translate the original annotation into the required format. Because there exist irregular quadrilaterals in certain datasets, we calculated the maximum circumscribed rectangle and regarded the long side as width and the short side as height. Then, the relative angle θ is defined as the angle between the width and the y-axis. For the width, height, and relative angle prediction, the Smooth-L_1 loss is adopted for L_{wh} and L_{angle} as follows:

$$\text{Smooth}L_1(x) = \begin{cases} 0.5 \times x^2, & \text{if} |x| < 1 \\ |x| - 0.5, & \text{otherwise.} \end{cases} \tag{10}$$

In (10), x means $d_p - d_g$, that is, the distance between the predicted value and true value. For the loss L_d of binary classification, the same loss setting as L_{ct} is adopted. Differently, the number of categories is modified referring to the position of the long side as described in Section III-C.

Finally, to learn the object instance, the class-wise binary instance mask is generated by transforming the quadrilateral. Then, the focal loss [70] is utilized to supervise training and the detail of L_m is as follows:

$$L_m = \frac{-1}{Z} \sum_{i}^{C} \begin{cases} \left(1 - \hat{M}_{xy}^i\right)^{\alpha} \log\left(\hat{M}_{xy}^i\right), & \text{if} M_{xy}^i = 1 \\ \left(\hat{M}_{xy}^i\right)^{\alpha} \log\left(1 - \hat{M}_{xy}^i\right), & \text{otherwise.} \end{cases} \tag{11}$$

In (11), $M \in [0, 1]^{W \times H}$, and Z denotes the number of positive samples (i.e., $M_{xy}^i = 1$).

IV. Experiments and Analysis

In this section, the following three contents are elaborated: 1) the experimental setup including dataset description, implementation details, and evaluation metrics; 2) comparison experiments and analysis; and 3) ablation experiments and analysis.

A. Experimental Setup

1) Dataset Description: To demonstrate the effectiveness of the proposed method, three public optical remote-sensing datasets DOTA [16], DIOR-R [31] and HRSC2016 [32] are introduced.

DOTA [16] dataset is a public large-scale benchmark dataset for object detection in aerial images, which collects 2806 images from different sensors and platforms. The original images range from 800 × 800 to about 4000 × 4.000 pixels and involve various spatial resolutions. The fully annotated dataset contains 188 282 instances and 15 categories [i.e., plane (P), baseball diamond (BD), bridge (B), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (S), tennis court (TC), basketball court (BC), storage tank (ST), soccer ball field (SBF), roundabout (R), harbor (HB), swimming pool (SP) and helicopter (H)], each of which is labeled by OBB. In this dataset, half of the original images are selected as the training set, 1/6 as the validation set, and 1/3 as the testing set. Here, all results involving the testing set are only obtained from the DOTA [16] online evaluation server.

DIOR-R [31] is a rotating version of DIOR [71]. It contains 23 463 images and 192 518 instances. The size of images is 800 × 800 pixels, and the spatial resolution ranges from 0.5 to 30 m. As a large-scale challenging object detection dataset, it provides 20 common object classes, including airplane

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(APL), airport (APO), baseball field (BF), BC, bridge (BR), chimney (CH), expressway serve area (ESA), expressway toll station (ETS), dam (DAM), golf field (GF), GTF, harbor (HA), overpass (OP), ship (SH), stadium (STA), ST, TC, train station (TS), vehicle (VE), and windmill (WM). For the data division, 11 725 images are selected as training and validation sets, and the remaining 11 738 images are employed for testing.

HRSC2016 [32] is a challenging ship detection dataset, which totally collects 1070 images from six famous harbors. In this dataset, there are a total of 2970 instances that are annotated by OBB. The image resolutions are between 0.4 and 2 m and the image size ranges from 700 to 1100 pixels. To ensure the distributions of the training set and testing set approximately match, the training, validation, and testing sets contain 436 images, 181 images, and 444 images, respectively.

2) Implementation Details: Our proposed detector regards CenterNet [20] as the baseline model. And then, ResNet101 [72] and ResNeSt101 [73] are used as the backbone network for comparing with other SOTA methods. ResNet50 [72] is used for evaluating the ablation experiments. During the training phase, all our models are trained for 140 epochs. The learning rate is initially set as 1.25e-4 and reduces by half at the 50th and 90th epochs. And, the Adam [74] optimizer is used in all experiments with batch size 10. Finally, the experimental environment is built on PyTorch 1.7.0 and two GeForce RTX 3090 graphic processing units (GPUs).

Because there exists a difference in the three datasets, they are processed separately. First, for the DOTA [16] dataset, following [18], [31] and considering limited computing resources, we adopted two processing strategies for ablation experiments and comparison experiments. For ablation experiments, we first resize original training/validation images into two scales $(1.0, 0.4)$ and then crop them into 640×640 patches by the overlap of 160. Random shifting and random flipping are applied to avoid overfitting. Notably, we use the training set for the training model and evaluate on the validation set. For comparison experiments, to be consistent with other SOTA methods, we reprocess the DOTA [16] dataset. Specifically, during the model training phase, we first resize the training and validation set into three scales $(1.3, 1.0, 0.4)$ and then split resized images into 640×640 with the overlap of 160. Moreover, we also introduce random rotating 0°, 90°, 180°, or 270° for data augmentation during training. In the inference, the testing set is resized to three scales $(1.5, 1.0, 0.5)$ and split into 1024×1024 patches with the overlap of 256. For the DIOR-R [31] dataset, we keep the original 800×800 image size for training and testing. For the HRSC2016 [32] dataset, we resize the images to 640×640 while keeping the aspect ratio. Then, the augmentation strategies of DIOR-R [31] and HRSC2016 [32] remain the same as DOTA [16], which are implemented for comparing with SOTA methods.

3) Evaluation Metrics: To measure the performance of different detectors, we adopt recall, precision, average precision (AP), and mean AP (mAP) defined in PASCAL visual object class criteria [75]. Notably, the intersection over union (IoU) between prediction and ground truth is set as 0.5 for evaluating performance. Among them, AP indicates the integral area under the precision and recall (P–R) curves. The specific description is as follows:

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{12}$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{13}$$

$$\text{mAP} = \frac{1}{C} \sum_{i=1}^{C} \text{AP}_i \tag{14}$$

where TP denotes the number of positive samples that are correctly identified, FP represents the number of negative samples that are incorrectly identified as positive, FN denotes the number of positive samples that are incorrectly identified as negative, and C represents the number of categories.

B. Comparison Analysis

In this section, extensive experiments are conducted on three public remote-sensing datasets (i.e., DOTA [16], DIOR-R [31], and HRSC2016 [32]) for comparing our proposed PIIDet with other SOTA detectors. And, these results are reported in Tables I– III, respectively, where our proposed PIIDet provides superior results, that is, 80.21% mAP on DOTA [16], 70.35% mAP on DIOR-R [31], and 90.64% AP on HRSC2016 [32]. In addition, the visualization of detection results is shown in Fig. 5. Furthermore, the detailed analysis is provided in the following three parts.

1) Results on the DOTA Dataset: The comparison on DOTA [16] is reported in Table I, which involves 15 anchor-based detectors and 13 anchor-free detectors. Specifically, different backbone networks and input image sizes are shown in the fourth and fifth columns of Table I. Regarding the inference efficiency, the frame per second (FPS) is calculated by detecting 40 pieces of 1024×1024 size images, as shown in the last column. Meanwhile, these methods that use multiscale training and testing strategies are marked as * in Table I.

According to Table I, one observation is that most anchor-based detectors can get better detection results than anchor-free detectors. We analyzed the reason is that due to the introduction of prior information being more intuitive, most researchers tended to focus on adopting well-designed preset anchors or CNN-based anchor refinement methods for better matching these oriented objects. Specifically, both SCRDet++ [6] and ReDet [12] set a large number of anchors per location of each pyramid to guarantee good performance, and their results can be seen in the ninth and 15th rows of Table I. In addition, ORPN [19] and S^2A-Net [18], respectively, designed oriented RPN and anchor refinement network in a two-stage detection framework for better making the anchors fit the oriented objects. Consequently, they, respectively, achieve the good performance of 80.52% and 79.15% mAP as shown in the 10th and 14th rows of Table I. We can conclude that the guidance information from prior anchors is very beneficial for the AOOD task. However, the prior anchors-based methods also brought extra hyperparameter settings and computational burden. It can be proved by the last column of Table I, which presents a decrease in inference efficiency compared with anchor-free methods. In pursuit of

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TABLE I

COMPARISON RESULTS ON DOTA [16]. * MEANS MULTISCALE TRAINING AND MULTISCALE TESTING. RED AND BLUE REPRESENT THE BEST AND THE SECOND BEST PERFORMANCE, RESPECTIVELY

		Method	Backbone	Input Size	mAP(%)	P	BD	B	GTF	SV	LV	S	TC	BC	ST	SBF	R	HB	SP	H	FPS
Anchor-based	Two-stage	RoI Trans.* [17]	ResNet101	1024	69.56	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	13.86
		FR-O [16]	ResNet101	1024	54.13	79.42	77.13	17.70	64.05	35.30	38.02	37.16	89.41	69.64	59.28	50.30	52.91	47.89	47.40	46.30	-
		CAD-Net [11]	ResNet101	1600	69.90	87.80	82.40	49.40	73.50	71.10	63.50	76.70	90.90	79.20	73.30	48.40	60.90	62.00	67.00	62.20	-
		Gliding Ver.* [27]	ResNet101	1024	75.02	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32	16.53
		MaskOBB [41]	ResNeXt101	1024	75.33	89.56	85.95	54.21	72.90	76.52	74.16	85.63	89.85	83.81	86.48	54.89	69.64	73.94	69.06	63.32	-
		CenterMap-Net* [76]	ResNet101	1024	76.03	89.83	84.41	54.60	70.25	77.66	78.32	87.19	90.66	84.89	85.27	56.46	69.23	71.13	71.56	66.06	-
		AMFFA-Net* [69]	ResNet101	1024	76.27	89.65	84.20	53.06	77.62	74.66	75.45	87.08	90.84	86.84	85.33	63.76	65.52	72.26	70.64	67.13	-
		FR-Est-FPN* [43]	ResNet50	1000	78.49	89.78	85.21	55.40	77.70	80.26	83.78	87.59	90.81	87.66	86.93	65.60	68.74	71.64	79.99	66.20	-
		ReDet* [12]	ReR50	1024	80.10	88.81	82.48	60.83	80.82	78.34	86.06	88.31	90.87	88.77	87.03	68.65	66.90	79.26	79.71	76.67	12.83
		ORPN* [19]	ResNet101	1024	80.52	90.26	84.74	62.01	80.42	79.04	85.07	88.52	90.85	87.24	87.96	72.26	70.03	82.93	78.46	68.05	15.89
	One-stage	CSL* [28]	ResNet152	800	76.17	90.25	85.53	54.64	75.31	70.44	73.51	77.62	90.84	86.15	86.69	69.60	68.04	73.83	71.10	68.93	11.95
		R^3Det* [7]	ResNet152	800	76.47	89.80	83.77	48.11	66.77	78.76	83.27	87.84	90.82	85.38	85.51	65.67	62.68	67.53	78.56	72.62	13.29
		KFIoU [14]	ResNet152	800	77.35	89.46	85.72	54.94	80.37	77.16	69.23	80.90	90.79	87.79	86.13	73.32	68.11	75.23	71.61	69.49	-
		S^2A-Net* [18]	ResNet101	1024	79.15	89.28	84.11	56.95	79.21	80.18	82.93	89.21	90.86	84.66	87.61	71.66	68.23	78.58	78.20	65.55	13.63
		SCRDet++* [6]	ResNet101	800	79.35	90.00	86.25	65.04	74.52	72.93	84.17	79.05	90.72	87.37	87.06	72.10	66.72	82.64	80.57	71.07	-
Anchor-free	Two-stage	APE [29]	ResNet50	1024	75.75	89.96	83.62	53.42	76.03	74.01	77.16	79.45	90.83	87.15	84.51	67.72	60.33	74.61	71.84	65.55	-
		TARDet* [25]	ResNet101	1024	79.57	89.70	85.41	58.28	79.55	78.24	85.54	89.04	90.68	85.76	86.33	69.03	70.70	82.16	73.37	69.86	-
		AOPG* [31]	ResNet101	1024	80.19	89.99	86.14	60.20	79.55	78.47	84.93	88.79	90.88	87.32	87.07	71.50	71.22	83.57	72.47	70.77	13.68
	One-stage	IENet* [24]	ResNet101	1024	57.14	80.20	64.54	39.80	32.07	49.71	65.01	52.58	81.45	44.66	78.51	46.54	56.73	64.40	64.24	36.75	-
		Axis learning. [42]	ResNet101	800	65.98	79.53	77.15	38.59	61.15	67.53	70.49	76.30	89.66	79.07	83.53	47.27	61.01	56.28	66.06	36.05	-
		O^2-Dnet* [49]	Hourglass-104	800	71.04	89.31	82.14	47.33	61.21	71.32	74.03	78.62	90.76	82.23	81.36	60.93	60.17	58.21	66.98	61.03	-
		SCPNet* [23]	Hourglass-104	768	72.20	88.18	78.30	48.32	63.18	72.52	75.86	87.73	90.78	80.15	85.95	55.53	63.30	67.11	60.74	65.39	-
		P-RSDet* [45]	Hourglass-104	1024	72.30	88.58	77.84	50.44	69.29	71.10	75.79	78.66	90.88	80.10	81.71	57.92	63.03	66.30	69.77	63.13	-
		DRN* [77]	Hourglass-104	1024	73.23	89.71	82.34	47.22	64.10	76.22	74.43	85.84	90.57	86.18	84.89	57.65	61.93	69.30	69.63	58.48	-
		BBAVector* [47]	ResNet101	600	75.36	88.63	84.06	52.13	69.56	78.26	80.40	88.06	90.87	87.23	86.39	56.11	65.62	67.10	72.08	63.96	20.31
		CBDA-Net* [46]	DLA34	800	75.74	89.17	85.92	50.28	65.02	77.72	82.32	87.89	90.48	86.47	85.90	66.85	66.48	67.41	71.33	62.89	-
		GGHL* [22]	DarkNet53	800	76.95	89.74	85.63	44.50	77.48	76.72	80.45	86.16	90.83	88.18	86.25	67.07	69.40	73.38	68.48	70.14	20.61
		Yu *et al*.* [50]	Res2Net50	800	77.18	89.84	84.90	51.55	74.42	78.54	83.28	87.36	90.73	87.17	85.70	65.20	62.23	75.56	72.21	69.01	-
		PIIDet* (Our)	ResNet101	640	78.38	89.09	84.83	53.44	70.16	79.52	84.69	88.16	90.80	88.04	87.38	68.14	66.51	76.14	72.34	76.38	23.08
		PIIDet* (Our)	ResNeSt101	640	80.21	89.67	86.04	55.76	70.64	81.35	85.85	88.40	90.79	88.37	88.16	66.91	72.40	77.74	81.95	79.05	16.61

TABLE II

COMPARISON RESULTS ON DIOR [31]. RED AND BLUE REPRESENT THE BEST AND THE SECOND BEST PERFORMANCE, RESPECTIVELY

	Method	Backbone	mAP(%)	APL	APO	BF	BC	BR	CH	DAM	ETS	ESA	GF	GTF	HA	OP	SH	STA	ST	TC	TS	VE	WM
Anchor-based	FR-O [16]	ResNet50	59.54	62.79	26.80	71.72	80.91	34.20	72.57	18.95	66.45	65.75	66.63	79.24	34.95	48.79	81.14	64.34	71.21	81.44	47.31	50.46	65.21
	Gliding Ver. [27]	ResNet50	57.55	61.49	28.52	73.57	81.17	23.98	72.54	19.94	72.39	58.20	69.25	79.54	32.14	44.87	77.71	67.57	61.09	81.46	47.33	38.01	60.24
	RoT Trans. [17]	ResNet50	63.87	63.34	37.88	71.78	87.53	40.68	72.60	26.86	78.71	68.09	68.96	82.74	47.71	55.61	81.21	78.23	70.26	81.61	54.86	43.27	65.52
	DODet [48]	ResNet50	65.10	63.40	43.35	72.11	81.32	43.12	72.59	33.32	78.77	70.84	74.15	75.47	48.00	59.31	85.41	74.04	71.56	81.52	55.47	51.86	66.40
Anchor-free	AOPG [31]	ResNet50	64.41	62.39	37.79	71.62	87.63	40.90	72.47	31.08	65.42	77.99	73.20	81.94	42.32	54.45	81.17	72.69	71.31	81.49	60.04	52.38	69.99
	Li *et al.* [26]	ResNet101	67.15	61.65	47.58	77.59	88.39	40.98	72.55	81.90	63.76	38.17	79.47	81.82	45.39	54.94	88.67	73.48	75.75	87.69	61.69	52.43	69.00
	PIIDet (Our)	ResNet101	67.35	77.90	31.84	79.37	89.99	44.96	72.72	30.74	79.16	80.06	76.59	81.28	44.26	57.94	89.04	69.28	73.10	89.50	56.96	59.87	62.39
	PIIDet (Our)	ResNeSt101	70.35	79.77	43.78	80.04	90.03	47.95	80.26	32.16	77.67	86.77	78.46	82.10	45.16	58.99	89.49	66.14	78.94	89.95	64.82	62.88	71.65

simplicity and efficiency, some researchers [23], [24], [42], [45], [49] have begun to get rid of the hand-designed rules and apply anchor-free algorithms on AOOD tasks. It can be observed that these methods cannot provide satisfactory results even if they designed novel OBB mapping parameters (i.e., the 19th–23th rows of Table I). On the contrary, as shown in the 18th and 26th rows of Table I, we found that both anchor-free oriented proposal generator (AOPG) [31] and CBDA-Net [46] achieve significant performance improvement than above-mentioned anchor-free methods by considering utilizing the supervised information from the output of network into network itself. This demonstrates that introducing learnable posterior information in these anchor-free methods helps improve the performance of AOOD.

From another perspective, although two-stage detectors such as AOPG [31], ORPN [19], and ReDet [12] have always dominated on the DOTA [16] dataset from the aspect of accuracy, multistage prediction scheme restricts the detection efficiency. From the 18th and 30th rows of Table I, it shows that under the anchor-free framework and when adapting the proposed OAPG and BCE-RegSpace, our proposed one-stage PIIDet not only has higher FPS than two-stage detector of AOPG [31], but also

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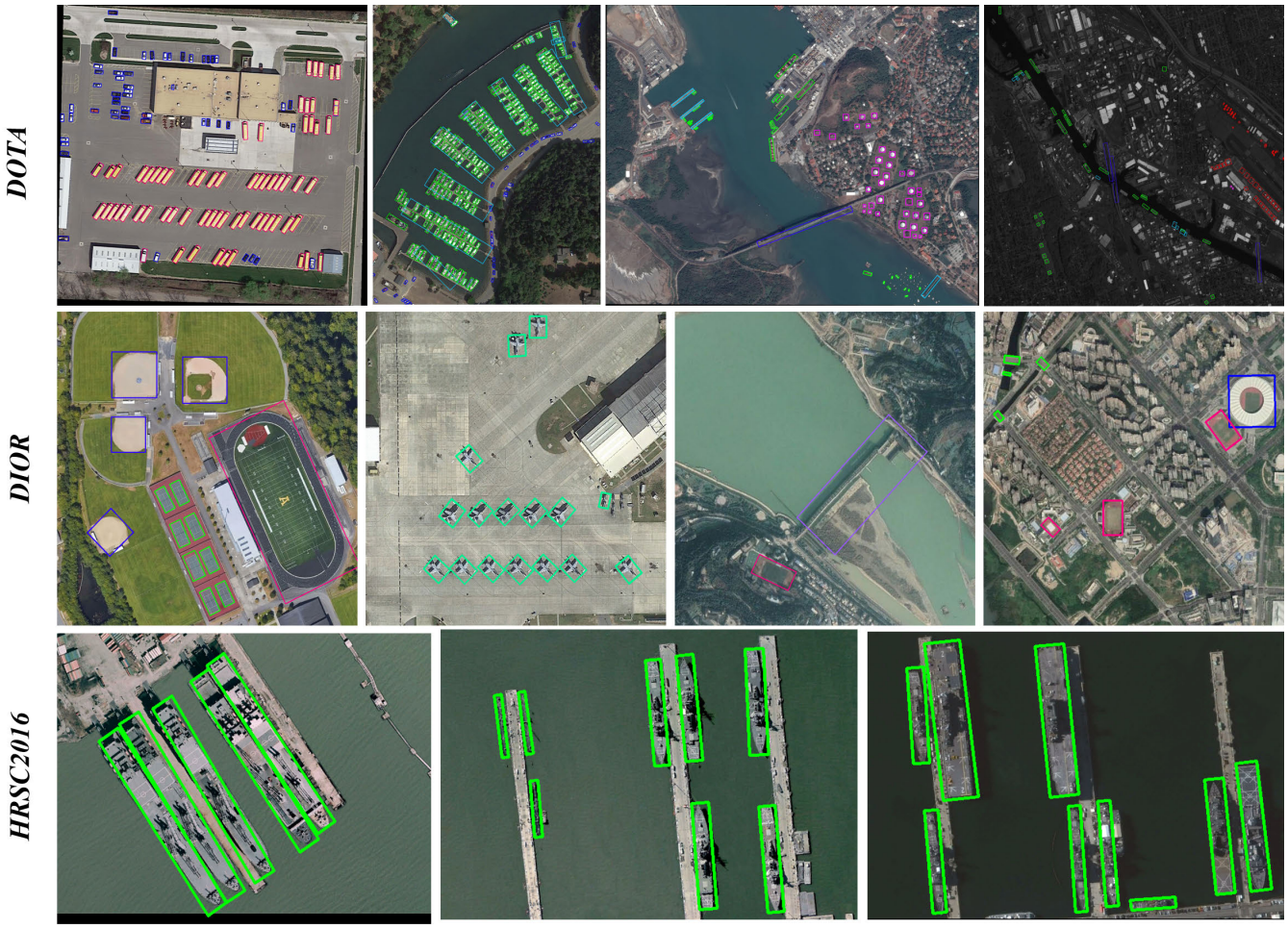


Fig. 5. Visualization of detection results obtained from the proposed PIIDet.

obtains the competitive result of 80.21% mAP. This shows that our proposed PIIDet can achieve the tradeoff between accuracy and efficiency. Moreover, from the comparison of these object categories with large aspect ratio [e.g., SV, LV, S and HB] in Table I, our proposed PIIDet has better detection performance by comparing with the fourth–seventh, 11th, 16th, 19th–23th, 25th rows of Table I which all design advanced mapping parameter of OBBs. This demonstrates our proposed PIIDet can also handle well-accurate angle prediction and provide precise OBBs. Finally, compared with all anchor-free one-stage detectors, our proposed PIIDet reaches the SOTA result of 80.21% mAP and also possesses a better inference speed of 23.08 FPS.

2) Results on the DIOR-R Dataset: As reported in Table II, the more challenging remote-sensing dataset DIOR-R [31] is utilized for evaluating our proposed PIIDet and other OBB detectors which include four anchor-based detectors and two anchor-free detectors. As illustrated in the first and third rows of Table II, FR-O [16] and Gliding Ver. [27] barely obtain 59.54% mAP and 57.55% mAP in DIOR [31] dataset, respectively, because not carefully considering the guidance information of prior anchors. Moreover, though some advanced methods such as RoI Trans. [17], DODet [48], AOPG [31], and Li et al. [26] have made some efforts on using guidance information and achieved pretty performance, our proposed PIIDet still outperforms them with 6.48%, 5.25%, 5.94%, and 3.20% mAP, which demonstrates the superiority of our proposed PIIDet. In addition, for some categories with large aspect ratios such as bridge (BR), vehicle (VE), and ship (SH), our proposed PIIDet also achieves the best performance, which demonstrates that our proposed PIIDet is a powerful and robust OBB detector on challenging the DIOR [31] dataset.

3) Results on the HRSC2016 Dataset: The HRSC2016 [32] is a ship detection dataset, which contains various ships with a large aspect ratio. Therefore, it allows further verification of the detection ability of remote-sensing objects that are sensitive to angle change. Here, nine anchor-based detectors and nine anchor-free detectors are introduced as a comparison. As shown in Table III, we found that the performance infimum of anchor-free detectors is lower than anchor-based detectors. We analyzed the reason that the anchor-based detectors generally can use the guidance information of prior anchors to quickly capture ships with arbitrary angles to achieve better performance. Furthermore, under the framework of an anchor-free one-stage detector that lacks guidance information, it can be found that when equipped with the designed OAPG and BCE-RegSpace, our proposed PIIDet can surpass other detectors and achieve the SOTA result of 90.64% AP.

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TABLE III

COMPARISON RESULTS ON HRSC2016 [32]. THE RESULTS ARE EVALUATED IN FORMAT OF PASCAL VOC 2007. RED AND BLUE REPRESENT THE BEST AND THE SECOND BEST PERFORMANCE, RESPECTIVELY

	Method	Backbone	Input Size	AP(%)@0.5
Anchor-based	AMFFA-Net [69]	ResNet101	~1333×800	89.90
	CSL [28]	ResNet50	800×800	89.62
	FR-Est-FPN [43]	ResNet50	1000×1000	89.70
	Gliding Ver. [27]	ResNet101	512×800	88.20
	ORPN [19]	ResNet101	1333×800	90.50
	R^3Net [7]	ResNet101	800×800	89.26
	RoI Trans. [17]	ResNet101	512×800	86.20
	S^2A-Net [18]	ResNet101	512×800	90.17
	ReDet [12]	ReR50	512×800	90.46
Anchor-free	AOPG [31]	ResNet50	800×1300	90.34
	Yu *et al.* [50]	Res2Net50	608×608	90.30
	CHPDNet [78]	Hourglass-104	512×512	90.60
	Axis learning [42]	ResNet101	800×800	78.15
	IENet [24]	ResNet101	1024×1024	75.01
	SCPNET [23]	Hourglass-104	768×768	89.10
	BBAvector [47]	ResNet101	608×608	88.60
	SKNet [79]	Hourglass-104	511×511	90.40
	GRS-Net [80]	ResNet101	800×800	89.57
	PIIDet (Our)	ResNeSt101	640×640	90.64

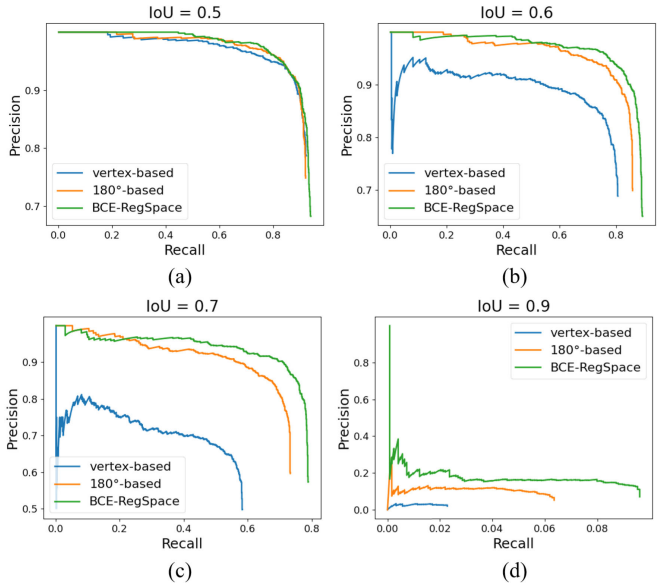


Fig. 6. Discussion about BCE-RegSpace on HRSC2016 [32]. (a) Comparison of P-R curves at the IoU of 0.5. (b) Comparison of P-R curves at the IoU of 0.6. (c) Comparison of P-R curves at the IoU of 0.7. (d) Comparison of P-R curves at the strictest IoU of 0.9.

C. Ablation Experiments

To demonstrate the effectiveness of these designed modules, we evaluated their performance on the DOTA [16] validation set. And, it should be noted that CenterNet [20] equipped with ResNet50 [73] is selected as the baseline model. The detailed experimental results are reported in Table IV, where these designed modules are gradually added to the baseline model to show their effects. From the first row of Table IV, it can be found that the baseline model combined with the widely used 180°-based angle regression method achieves 70.91% mAP.

TABLE IV

ABLATION EXPERIMENTS ON DOTA [16]

Method	Module					mAP(%)@0.5
	Angle	BCE-RegSpace	OAPG		MSRL	
			HFFM	OA-PII		
Baseline	✓					70.91
		✓				71.64(+0.73)
		✓	✓			72.03(+1.12)
		✓	✓	✓		72.45(+1.54)
PIIDet		✓	✓	✓	✓	**73.23**(+2.32)

TABLE V

WEIGHT HYPERPARAMETER DISCUSSION OF BCE-REGSPACE

Dataset	λ_{wh}	$\lambda_a : \lambda_d$			
		1:1	0.5:1	0.2:1	0.1:1
DOTA [16]	1	70.40	69.62	**71.64**	70.48
	0.1	69.67	69.95	**71.42**	70.40
HRSC2016 [33]	1	89.02	89.33	**89.53**	89.35
	0.1	88.01	88.24	**89.13**	87.39

* mAP(%) for DOTA and AP(%) for HRSC2016.

Then, when replacing the 180°-based angle regression method with the proposed BCE-RegSpace, it obtains 0.73% mAP gain, which proves the BCE-RegSpace can supply more suitable angle regression space and promote more accurate prediction of OBBs. Furthermore, based on the proposed BCE-RegSpace, the effectiveness of the OAPG structure including HFFM and OA-PII is further verified. First, as shown in the third row of Table IV, the result shows that the designed HFFM applied in the BCE-RegSpace-based detector can bring 1.12% mAP gain over the baseline, which indicates the HFFM can provide effective multiscale semantic feature description. Next, as reported in the fourth row of Table IV, when the designed OA-PII module is further incorporated into above network, it gains 1.54% mAP over the baseline, which demonstrates the OA-PII module can condition the learnable posterior instances into the prediction of OBB mapping parameters of BCE-RegSpace to achieve the effective posterior guidance. Finally, to be compatible with multiscale remote-sensing objects and make full use of the multiscale semantic mapping space obtained from HFFM, the MSRLs are constructed. As reported in the fifth row of Table IV, it can be observed that the complete PIIDet including BCE-RegSpace, OAPG, and MSRL can achieve 2.32% mAP gain compared to the baseline model. In general, according to the above ablation analysis, these designed modules in the PIIDet are very beneficial for AOOD. In the following content, more ablation analysis will be further elaborated.

1) Ablation Analysis on BCE-RegSpace: To further study the improvement of angle prediction by using BCE-RegSpace, we utilized stricter IoUs to calculate AP on the HRSC2016 [32] dataset. Notably, HRSC2016 [32] contains a lot of large aspect ratio ships, thus it is more sensitive for angle regression. In detail, four different IoUs, that is, 0.5, 0.6, 0.7, and 0.9, are chosen to compare with other two classical OBB mapping parameters, that is, 180°-based angle regression method [39] and vertex-based method [27]. Here, to ensure fairness,

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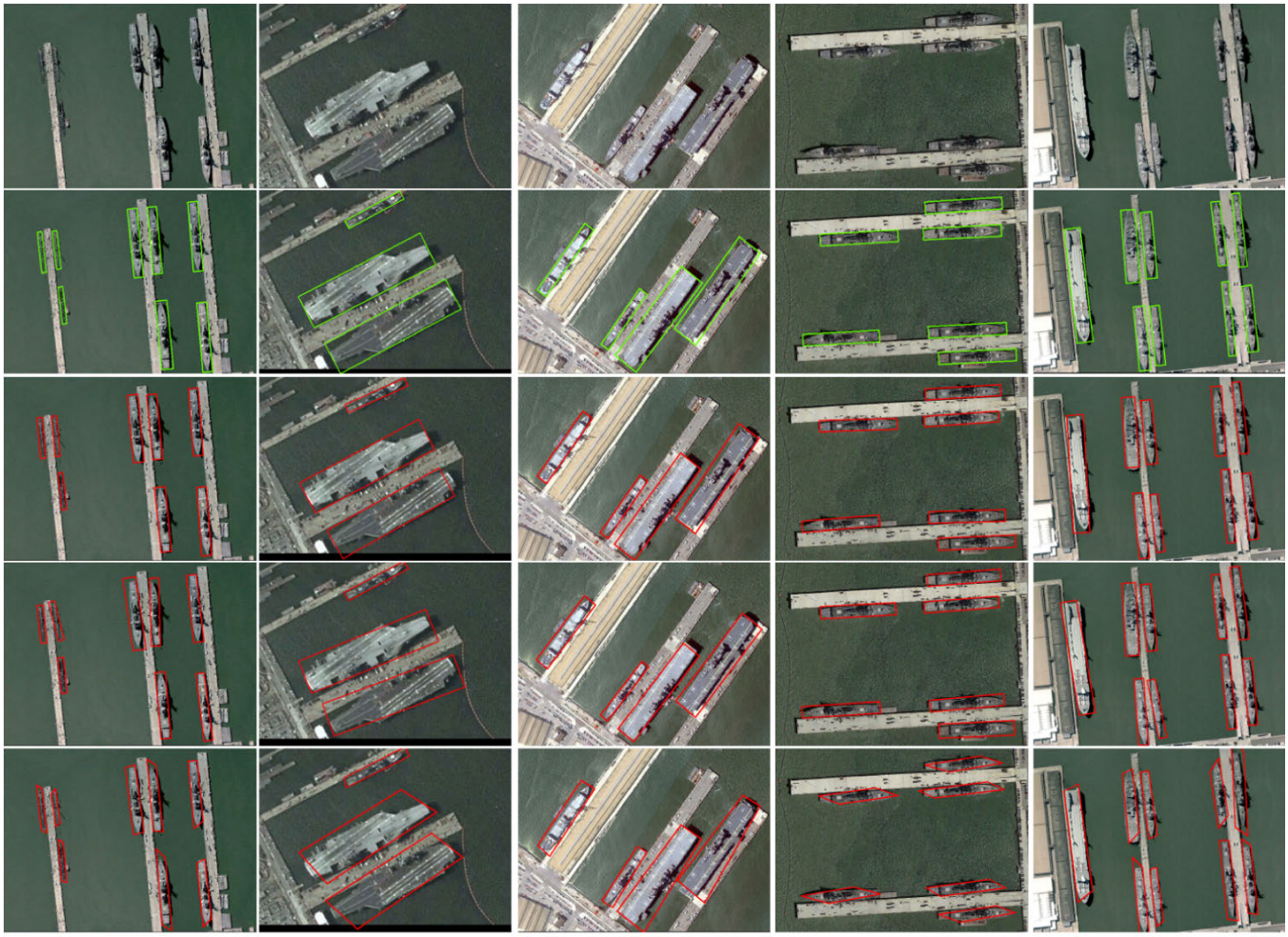


Fig. 7. Comparison results of three different OBB mapping parameters. Among them, the first row denotes original images. The second row denotes the ground truths. The third row denotes the detection results from the proposed BCE-RegSpace. The fourth row denotes the detection results from 180°-based angle regression method. The fifth row denotes the detection results from vertex-based method. Moreover, (a)–(e) indicate the five harbor scenarios in HRSC2016 [32].

TABLE VI
COMPARISON WITH DIFFERENT PRECONDITIONS ON THE OAPG STRUCTURE

Method	Pre-Condition	mAP(%)@0.5
Baseline + BCE-RegSpace	ResNet50+TransConv [20]	70.54
	DLA-34 [81]	70.68
	HRNet [82]	72.17
	Hourglass-104 [83]	71.31
	ResNet50+FPN [84]	71.24
	ResNet50+BiFPN [85]	70.92
	ResNet50+HFFM	**72.34**

* The single-scale regression layer is used for evaluation.

TABLE VII
COMPARISON WITH DIFFERENT POSTERIOR INJECTION METHODS ON THE OAPG STRUCTURE

Method	Posterior Injection Method	mAP(%)@0.5
Baseline + BCE-RegSpace	OCSA	71.75
	CWSA	72.33
	CWSCA	**72.34**

* The single-scale regression layer is used for evaluation.

we only modify the head network and postprocessing of the baseline model according to three different OBB mapping parameters, and other network structures remain unchanged. The results are reported in Fig. 6. One observation from Fig. 6 is that the designed BCE-RegSpace maintains the best AP in four IoUs setting. Meanwhile, with the increase of IoU, the advantage of our designed BCE-RegSpace is gradually more significant. In addition, under the IoU settings of 0.5 and 0.6, the recall has a relatively small gap among them, but when reaching the IoU settings of 0.7 and 0.9, the recall of BCE-RegSpace exceeds other methods more obviously, which indicates that our proposed BCE-RegSpace can predict more accurate OBBs. Then, by intuitive analysis, Fig. 7 shows the detection results of three different OBB mapping parameters. It can be found from the fourth row of Fig. 7, the 180°-based angle regression method produces some OBBs with inaccurate angles. From the fifth row of Fig. 7, although these OBBs generated from vertex-based methods can cover

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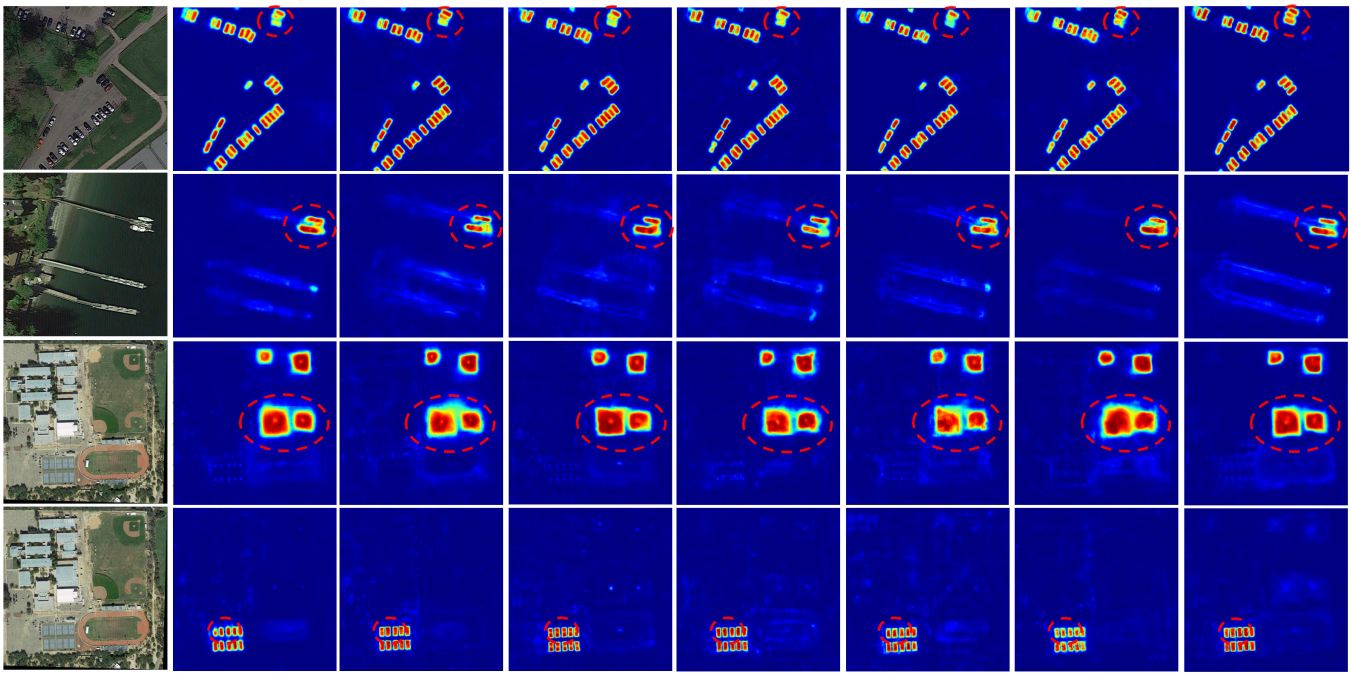


Fig. 8. Instance prediction results from different preconditions of the OAPG structure. (a) Original image. (b) ResNet50+TransConv. (c) DLA34. (d) HRNet. (e) Hourglass-104. (f) ResNet50+FPN. (g) ResNet50+BiFPN. (h) Our proposed ResNet50+HFFM.

these ships, the quality of OBBs is poor. In a word, the above comparison results both demonstrated that the designed BCE-RegSpace mapping parameter can provide a more accurate OBB description and promote performance gain.

In addition, due to the BCE-RegSpace mapping parameter including two prediction branches (i.e., binary classification and relative angle), the weight ratio between these two branches is further analyzed. It is worth noting that the weight of binary classification is first set as 1 and then adjust the weight of relative angle prediction to 1, 0.5, 0.2, and 0.1. Besides, since the regression of width and height would affect the selection of the long side, λ_{wh} is also introduced to discuss. As reported in Table V, for DOTA [16] and HRSC2016 [32], the best results are both obtained under the $\lambda_a : \lambda_d$ weight ratio of 0.2:1.

2) Ablation Analysis on OAPG: In this section, the OAPG structure is further discussed. It is worth noting that a plain network structure with a single-scale regression layer is employed to clearly verify its effectiveness. Additionally, λ_{wh}, λ_a, and λ_d are set as 1, 0.1, and 1, respectively. Based on the above conditions, the ablation analysis is conducted. First, in Section III, the importance of powerful multiscale semantic feature mapping space for ensuring the quality of learnable instance prediction has been emphasized. To prove this, six different feature fusion methods such as ResNet50+Transposed Convolution (TransConv) [20], DLA-34 [81], HRNet [82], Hourglass-104 [83], ResNet50+FPN [84], and ResNet50+BiFPN [85] are used as precondition of OAPG to compare. The results are reported in Table VI. When adopting the plain feature fusion method, that is, ResNet50+TransConv, it achieves the worst 70.54% mAP on the DOTA [16] validation set. With the replacement of other feature fusion methods, they provide varying degrees of performance gain. In the last row of Table VI, it is evident that when combined with HFFM, the OAPG can reach the best performance of 72.34% mAP, which presented that powerful multiscale semantic feature mapping space is key in our designed OAPG structure. In addition, we visualized the instance prediction results to compare different feature fusion methods in Fig. 8. Compared with columns (b)–(h) in Fig. 8, it can be seen that column (h) in Fig. 8 obtained from ResNet50+HFFM has clearer instance shape description of the objects with different categories, as shown in the red circle area of Fig. 8. Through this quantitative and intuitive analysis, it can be concluded that our proposed HFFM can provide powerful multiscale semantic feature mapping space and endow the OAPG with enough accurate guidance conditions for the posterior guiding process.

In addition, some studies [6], [15], [69] have verified the effectiveness of posterior injected information by supervised learning for AOOD. To verify the superiority of our posterior injection method, different injection methods are compared in Table VII. Here, one-channel spatial attention (OCSA) denotes that the attention maps are obtained by only focusing on foreground or background regions. Then, class-wise spatial attention (CWSA) denotes that it can distinguish not only foreground and background, but also different categories. In our posterior injection method of OAPG structure, we adopt the way of CWSA, and further reweight the CWSA by class channel attention, called class-wise spatial and channel attention (CWSCA). As reported in Table VII, the experimental results show that using OCSA (i.e., the first row) performs lower mAP than CWSA and CWSCA (i.e., the second and third rows). The reason is that the OCSA cannot describe

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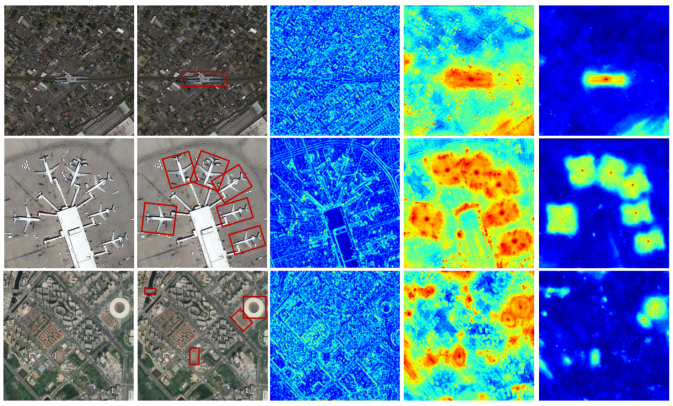


Fig. 9. Visualization result of feature maps from different phases in the proposed OAPG. (a) Original image. (b) Foreground objects labeled by red OBBs. (c) Feature maps before entering into the OAPG structure. (d) Feature maps from the outputs of the HFFM module. (e) Feature maps from the outputs of the entire OAPG. Notably, the red points located in (e) represent the center points of objects.

the shape and orientation of objects with different categories well, especially for the dense arranged or covered objects. When decoupling foreground objects by category, the dense objects with different categories (e.g., ship and harbor) can be allocated into independent channels to keep the shape and orientation information. In addition, considering that not all categories appear in a single image, the class channel attention is introduced based on CWSA. By comparing the second and third rows, it can be found that CWSCA would bring more gain, which proved that further enhancing the objects of the existing category is beneficial in our method.

Finally, from a holistic perspective, the visualized feature maps are presented in Fig. 9 to visually observe the changes in feature maps through the entire OAPG structure. Here, some complex remote-sensing scenes are selected, as shown in Fig. 9(a). Notably, the drawing approach of these visualization results is to first perform a sum operation along the channel dimension of feature maps and then draw a heat map along the spatial dimension. According to this operation, first, as shown in Fig. 9(c), three visualization results are generated from the feature maps before entering the OAPG. It is evident that the background noises spread out the entire feature map, and the foreground objects are difficult to distinguish. Then, as shown in Fig. 9(d), three visualization results are obtained from feature maps that pass through the HFFM of the OAPG. It can be observed that the foreground objects have more obvious feature responses to easily distinguish them from background noises. However, considerable noises from background regions are still around foreground objects. Finally, as shown in Fig. 9(e), three visualization results are generated from the feature maps that are processed by the entire OAPG structure. It can be observed that after injecting the high-quality posterior instance guidance, the feature response of object regions in these feature maps performs more prominently. Meanwhile, the injected posterior instance guidance also helps understand the interpretation of the center point, as shown in the red regions of Fig. 9(e). Moreover, compared with Fig. 9(c) and (d), the complex background interference is almost completely suppressed. To summarize, the proposed OAPG structure is a very effective posterior guidance approach for mapping multicategory and arbitrary-oriented remote-sensing objects in complex backgrounds.

V. Conclusion

In this article, a novel one-stage detector called PIIDet is proposed to dramatically promote the performance of AOOD in an anchor-free manner, and it consists of OAPG structure and BCE-RegSpace. Concretely, based on the idea of posterior guidance, a class-wise instance prediction branch is introduced in the OAPG structure, which is driven by a carefully designed HFFM. Then, a class-wise and spatial-wise posterior injection method is designed for conditioning the learning of the proposed OBB mapping parameters on the learned instance description coefficients. Second, a concise, yet effective BCE-RegSpace-based OBB mapping parameter is proposed, and it can reasonably tackle the boundary discontinuity issue of the angular period and provide accurate OBB prediction. Finally, extensive experiments are conducted on DOTA [16], DIOR-R [31], and HRSC2016 [32] datasets. The comparison results and detailed ablation results indicate that the proposed PIIDet can obtain the advanced performance of other AOOD detectors. Furthermore, we will further exploit the guidance mechanism for improving the performance of the AOOD task.

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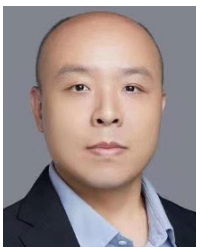
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